

BrighterSQL Exercises 5 - SQL Date Functions - Ivy Charles

Of QUESTION 1-5 Extract Dates Parts

QUESTION 1

```
SELECT  
    order_id,  
    customer_id,  
    order_date,  
    DAYNAME(order_date) AS day-name  
FROM orders;
```

order_id	customer id	order_date	day-name
1001	101	2026-05-01	Friday
1002	102	2026-05-02	Saturday
1003	103	2026-05-03	Sunday
1004	104	2026-05-04	Monday
1005	105	2026-05-05	Tuesday

QUESTION 2

SELECT customer_id,
customer_name,
Sign up date,
MONTH NAME(Signup date) AS signup-month-name
FROM Customer-signups;

customer_id	customer_name	Sign up date	signup-month-name
201	John	2026-01-15	January
202	Mary	2026-02-20	February
203	Peter	2026-03-05	March
204	Sarah	2026-04-18	April
205	Thabo	2026-05-30	May

QUESTION 3

SELECT sale_id,
Product-name,
Sale-date,
EXTRACT(MONTH FROM sale-date) AS sale-month
FROM sales;

Sale_id	Product-name	sale-date	sale-month
S001	Laptop	2026-01-10	1
S002	Mouse	2026-02-15	2
S003	Keyboard	2026-03-20	3
S004	Monitor	2026-04-25	4
S005	Desk	2026-05-30	5

QUESTION 4

SELECT transaction-id,
customer-id,
transaction-date,
EXTRACT (YEAR FROM transaction-date) AS
transaction-year

FROM transactions;

transaction-id	customer-id	transaction-date	transaction-year
T001	101	2024-12-15	2024
T002	102	2025-01-20	2025
T003	103	2025-06-10	2025
T004	104	2026-02-05	2026
T005	105	2026-05-25	2026

QUESTION 5

SELECT delivery-id,
customer-id,
delivery-date,
EXTRACT (DAY FROM delivery-date) AS
day-of-month

FROM deliveries;

delivery-id	customer-id	delivery-date	day-of-month
D001	101	2026-05-01	01
D002	102	2026-05-08	08
D003	103	2026-05-15	15
D004	104	2026-05-22	22
D005	105	2026-05-29	29

Q2 QUESTION 6-10 → CURRENT DATE, convert & Format.

QUESTION 6

SELECT employee_id,
employee_name,
department,
CURRENT_DATE() AS today_date
FROM employees;

employee_id	employee_name	department	today_date
301	Mandi	Sales	2026-07-30
302	Brian	IT	2026-07-30
303	Lerato	Finance	2026-07-30
304	Sipho	HR	2026-07-30
305	Aisha	Marketing	2026-07-30

QUESTION 7

SELECT order_id,
customer_id,
order_date_text,
TO_DATE (order_date_text, 'YYYYMMDD') AS order_date
FROM online_orders;

order_id	customer_id	order_date_text	order_date
4001	101	2026-01-15	2026-01-15
4002	102	2026-02-20	2026-02-20
4003	103	2026-03-25	2026-03-25
4004	104	2026-04-10	2026-04-10
4005	105	2026-05-05	2026-05-05

QUESTION 8

SELECT payment_id,
customer_id,
payment_date,

TO_CHAR (payment_date, 'YYYY MM DD') AS

Formatted_payment_date.

FROM payment_dates;

payment_id	customer_id	payment_date	Formatted_payment_date
P001	101	2026-01-05	2026-01-05
P002	102	2026-02-10	2026-02-10
P003	103	2026-03-15	2026-03-15
P004	104	2026-04-20	2026-04-20
P005	105	2026-05-25	2026-05-25

QUESTION 9

SELECT customer_id,

customer_name,

last_purchase_date,

DATE_DIFF (CURRENT_DATE (), last_purchase_date)

AS days_since_last_purchase.

FROM customer_purchases

customer_id	customer_name	last_purchase_date	days_since_last_purchase
S01	John	2026-05-01	90
S02	Mary	2026-05-10	81
S03	Peter	2026-05-15	76
S04	Sarah	2026-05-20	71
S05	Thabo	2026-05-25	66

QUESTION 10

SELECT *

DATEADD(DAY, 7, order-date) AS expected-delivery-date

FROM shipping-address

order-id	customer-id	order-date	expected-delivery-date
6001	101	2026-05-01	2026-05-08
6002	102	2026-05-03	2026-05-10
6003	103	2026-05-05	2026-05-12
6004	104	2026-05-07	2026-05-14
6005	105	2026-05-09	2026-05-16

03 QUESTION 11-15 - Multi-Extract & Date Filtering,

QUESTION 11

~~SELECT~~ booking-id,
customer-id,
booking-date,

YEAR(booking-date) AS booking-year
MONTH(booking-date) AS booking-month
DAY(booking-date) AS booking-day

FROM bookings

booking-id	customer-id	booking-date	booking-year	booking-month	booking-day
B001	101	2026-01-12	2026	1	12
B002	102	2026-02-18	2026	2	18
B003	103	2026-03-22	2026	3	22
B004	104	2026-04-09	2026	4	9
B005	105	2026-05-27	2026	5	27

QUESTION 12

SELECT order_id,
customer_id,
order_date,

EXTRACT (YEAR FROM order_date) AS order_year,

amount

FROM yearly-orders

WHERE EXTRACT (YEAR FROM order_date) = 2026;

order_id	customer_id	order_date	order_year	amount
7004	104	2026-02-05	2026	1500
7005	105	2026-05-25	2026	2000

QUESTION 13

SELECT order_id,
customer_id,
order_date,

MONTH (order_date) AS order_month,

amount

FROM monthly-orders

WHERE MONTH (order_date) = 3;

order_id	customer_id	order_date	order_month	amount
8003	103	2026-03-10	3	800
8004	104	2026-03-25	3	1500

QUESTION 14

SELECT subscption_id,
customer_id,
start_date,

LAST_DAY (start_date) AS month_end_date
FROM subscriptions;

subscption_id	customer_id	start_date	month_end_date
SUB001	101	2026-01-10	2026-01-31
SUB002	102	2026-02-15	2026-02-28
SUB003	103	2026-03-20	2026-03-31
SUB004	104	2026-04-25	2026-04-30
SUB005	105	2026-05-30	2026-05-31

QUESTION 15

SELECT send_id,
customer_id,
send_date,

DATE_TRUNC ('MONTH', send_date) AS month_start_date

FROM campaign_sends;

send_id	customer_id	send_date	month_start_date
C001	101	2026-01-12	2026-01-01
C002	102	2026-02-13	2026-02-01
C003	103	2026-03-22	2026-03-01
C004	104	2026-04-09	2026-04-01
C005	105	2026-05-27	2026-05-01

04 QUESTION 16-20 - Advanced date work

QUESTION 16

SELECT invoice_id,
customer_id,
invoice_date,
TO_CHAR (invoice_date, 'Month YYYY') AS
invoice_month_year

FROM invoice_dates;

invoice_id	customer_id	invoice_date	invoice_month_year
INV001	101	2026-01-05	January 2026
INV002	102	2026-02-10	February 2026
INV003	103	2026-03-15	March 2026
INV004	104	2026-04-20	April 2026
INV005	105	2026-05-25	May 2026

QUESTION 17

SELECT customer_id,
customer_name,
date_of_birth,
DATEDIFF (YEAR, date_of_birth, '2026-07-30') AS
customer_age
FROM customer_birthdays;

customer_id	customer_name	date_of_birth	customer_age
901	John	1998-05-10	28
902	Mary	1990-08-20	35
903	Peter	2002-03-15	24
904	Sarah	1985-12-01	40
905	Theo	2000-07-30	26

QUESTION 18

```
SELECT order_id,  
       customer_id,  
       order_date,  
       DAYNAME (order_date) AS day_name,  
       CASE  
         WHEN DAYNAME (order_date) IN ('Saturday', 'Sunday')  
         THEN 'weekend'  
         ELSE 'weekday'  
       END AS day_type  
FROM weekend_orders;
```

order_id	customer_id	order_date	day_name	day_type
9001	101	2026-05-01	Friday	weekday
9002	102	2026-05-02	Saturday	weekend
9003	103	2026-05-03	Sunday	weekend
9004	104	2026-05-04	Monday	weekday
9005	105	2026-05-05	Tuesday	weekday

QUESTION 19

```
SELECT transaction_id, customer_id, transaction_date,  
       QUARTER (transaction_date) AS transaction_quarter,  
       amount  
FROM quarterly_transactions;
```

transaction_id	customer_id	transaction_date	transaction_quarter	amount
Q001	101	2026-01-15	1	500
Q002	102	2026-03-20	1	1200
Q003	103	2026-04-10	2	800
Q004	104	2026-07-05	3	1500
Q005	105	2026-10-25	4	2000

QUESTION 20

SELECT order_id,
customer_id,
order_date,

DATEDIFF ('2026-07-30', order_date) AS
days_since_order,

amount

FROM recent_orders

WHERE DATEDIFF ('2026-07-30', order_date) > 30;

order_id	customer_id	order_date	days_since_order	amount
R001	101	2026-04-01	120	500
R002	102	2026-04-15	106	1200
R003	103	2026-05-01	90	800
R004	104	2026-05-10	81	1500
R005	105	2026-05-25	66	2000